NeuroPulse AI: Electromyography (EMG) Profile

Yellow – cleaned signal | Green – muscle strength | Red – muscle active shadow

Patient: Patient Name | **ID:** NP-20260522-6079 | **Date:** 22-05-2026 17:46

Avg Strength	Active Time	Major Spikes	Signal Quality
3.4	**3.9%**	**4**	**Low**

1. Clinical Summary

Session Score	63/100
Symmetry Score	83.8%
Side Deficit	16.2% (Left lower)
Lowest Activity Channel	Channel 1
Fatigue Drop	Low (0.0% decline)
Signal Quality	Low

2. 30-Second Graph Snapshot

3. Muscle-wise Quantitative Summary

Muscle / Channel	Avg	Peak	Active %	Quality
Channel 1	0.7	71.6	1.3%	Low
Channel 2	5.5	293.6	5.3%	Low
Channel 3	2.7	161.3	3.9%	Low
Channel 4	4.7	331.5	4.9%	Low

4. Local Data-Based EMG Findings

■ **Signal Reasoning (Quantitative Text Analysis):**

* Session Score: 38/100. Signal Quality: Low.
* Side Deficit: 68.8%. Symmetry Score: 31.2%. Marked left-side deficit detected.
* Lowest Activity: Channel 1. Highest Peak: Channel 4.

* Fatigue Drop: Low (0.0% decline from early to recent activity).
* Channel Balance: left-side channel balance 0%; right-side channel balance 0%.
* Suggested focus: compare the two right-side muscles/channels and train the weaker activation pattern.
* Selected affected side: Select Affected Side. Non-diagnostic rehab monitoring support only.

Recorded Session Findings:

* Recording summary: 6850 EMG samples captured over approximately 60 seconds.
* Session Score: 63/100. Signal Quality: Low.
* Side Deficit: 16.2%. Symmetry Score: 83.8%. Impression: mild to moderate left-side activation deficit. Side-to-side activation difference is present and should be monitored.
* Lowest Activity: Channel 1 showed the lowest average activation (0.7).
* Highest Peak: Channel 4 showed the highest peak activation (331.5).
* Channel Balance: left-side channel balance 12.0%; right-side channel balance 56.3%.
* Fatigue Drop: Low fatigue pattern with 0.0% decline from early to recent activity.
* Sustained activation: Channel 1 1.3%, Channel 2 5.3%, Channel 3 3.9%, Channel 4 4.9% above threshold.
* Clinical note: Affected side was not selected; interpretation is limited to channel activity and symmetry. This is a non-diagnostic rehab monitoring prototype that provides quantitative EMG feedback for physiotherapy support.

5. Clinical Observations & Notes

Summary Notes: N/A

Comparative Analysis:

• Left Side Avg: 3.1
• Right Side Avg: 3.7
• Symmetry Ratio (L/R): 83.8%
• Most Dynamic Channel: Channel 4